

Complexity Analysis: ϵ vs. δ in Hoeffding's Inequality

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Problem Statement

Starting from the multi-hypothesis Hoeffding bound equation:

$$\delta = 2Me^{-2N\epsilon^2}$$

determine which parameter (ϵ or δ) controls the required sample size N more efficiently, where N represents the number of data points.

1. Algebraic Derivation of N

To evaluate the effect of each parameter on N , we solve for N step by step:

1. **Isolate the exponential term:**

$$\frac{\delta}{2M} = e^{-2N\epsilon^2}$$

2. **Apply the natural logarithm to both sides:**

$$\ln\left(\frac{\delta}{2M}\right) = -2N\epsilon^2$$

3. **Invert the fraction to remove the negative sign:**

$$\ln\left(\frac{2M}{\delta}\right) = 2N\epsilon^2$$

4. **Solve for N :**

$$N = \frac{1}{2\epsilon^2} \ln\left(\frac{2M}{\delta}\right)$$

2. Parameter Sensitivity Analysis

By examining the derived expression for N , we observe distinct functional dependencies on ϵ and δ :

A. Dependence on Error Tolerance (ϵ)

The sample size N scales inversely with the square of the precision parameter:

$$N \propto \frac{1}{\epsilon^2}$$

Impact: This polynomial (quadratic) dependency implies that increasing accuracy is computationally expensive. For instance, halving the error tolerance ϵ (e.g., from 0.1 to 0.05) requires a **fourfold increase** in the required sample size N .

B. Dependence on Failure Probability (δ)

The sample size N scales logarithmically with the confidence parameter:

$$N \propto \ln\left(\frac{1}{\delta}\right)$$

Impact: Due to the logarithmic growth, requesting a significantly higher confidence level is remarkably cheap. For example, reducing the failure probability δ by a factor of 1,000 (e.g., from 10% to 0.01%) increases N only by an additive factor proportional to $\ln(1000) \approx 6.91$.

Conclusion

The confidence parameter δ **controls** N **much more efficiently than** ϵ .

Because of the logarithmic relationship, high confidence ($\delta \rightarrow 0$) can be achieved with a minimal increase in sample size. In contrast, the quadratic dependence on the error margin ϵ makes ϵ the primary bottleneck governing data requirements.